

Bias, Variance in Machine Learning

CS60050: Machine Learning

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Agenda

- § Understand what is bias and variance in a machine learning model
- § Understand their relation to overfitting and underfitting and the interplay between them

Resources

- § The course “Learning from Data” by Yaser Abu-Mostafa [Lecture 8].
- § The Book “Learning from Data” by Yaser Abu-Mostafa, Malik Magdon-Ismail and Hsuan-Tien Lin.

(Recap): Machine Learning and Optimization

- § In learning we care about some performance measure P (e.g., image classification accuracy, language translation accuracy etc.) on **test set**, but we minimize a different cost function $J(\theta)$ on **training set**, with the hope that doing so will improve P .
- § This is in contrast to pure optimization where minimizing $J(\theta)$ is a goal in itself.
- § Lets see what types of errors creep in as a result

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Expected and Empirical Risk

- § Training data is $\{\mathbf{x}^{(i)}, y^{(i)} : i = 1, \dots, n\}$ coming from probability distribution $\mathcal{D}(\mathbf{x}, y)$.
- § Let the learned hypothesis function and the loss are denoted as $g^{\mathcal{D}}(\mathbf{x})$ and $l(g^{\mathcal{D}}(\mathbf{x}), y)$ respectively.
- § What we want to minimize is the expected risk:

$$E(g^{\mathcal{D}}(\mathbf{x})) = \int l(g^{\mathcal{D}}(x), y) d\mathcal{D}(x, y) = \mathbb{E}(l(g^{\mathcal{D}}(x), y))$$

- § If we could minimize this risk we would have got the true optimal function

$$f = \arg \min_{g^{\mathcal{D}}} E(g^{\mathcal{D}})$$

- § But we don't know the actual distribution that generates the data. So, what we actually minimize is the empirical risk:

$$E_n(g^{\mathcal{D}}) = \frac{1}{N} \sum_{i=1}^N l(g^{\mathcal{D}}(x_i), y_i) = \mathbb{E}_n(l(g^{\mathcal{D}}(x), y))$$

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Expected and Empirical Risk

- § We choose a family $\mathcal{H}$ of candidate prediction functions and find the function that minimizes the empirical risk:

$$g_n^{\mathcal{D}} = \arg \min_{g^{\mathcal{D}} \in \mathcal{H}} E_n(g^{\mathcal{D}})$$

- § f may not be found by minimizing empirical risk, we also define:

$$g_{\mathcal{H}^*}^{\mathcal{D}} = \arg \min_{g^{\mathcal{D}} \in \mathcal{H}} E(g^{\mathcal{D}})$$

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	Minimizes	Staying within	
f	expected risk	No constraints on function family	• True data distribution known • Family of functions exhaustive
$g_{\mathcal{H}^*}^{\mathcal{D}}$	expected risk	family of functions $\mathcal{H}$	• True data distribution known • Family of functions not exhaustive
$g_n^{\mathcal{D}}$	empirical risk	family of functions $\mathcal{H}$	• True data distribution not known • Family of functions not exhaustive

§ True data generation procedure not known

§ Family of functions [or hypothesis] to try is not exhaustive

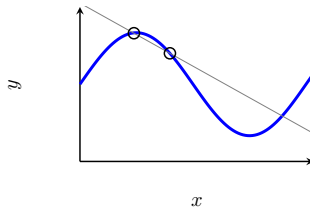
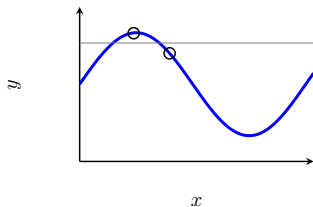
§ (And not to forget) we rely on a surrogate loss in place of the true classification error rate

A Simple Learning Problem

2 Data Points. 2 hypothesis sets:

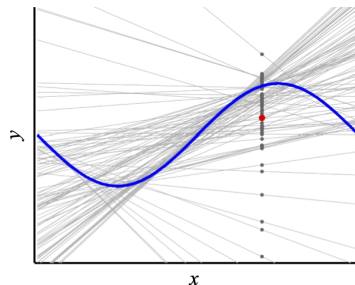
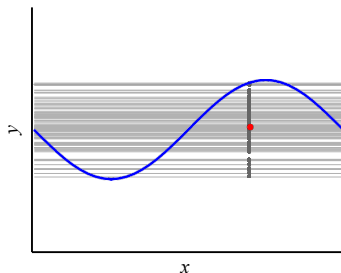
§ $\mathcal{H}_0 : h(x) = b$

§ $\mathcal{H}_1 : h(x) = ax + b$



Slide courtesy: Malik Magdon-Ismail

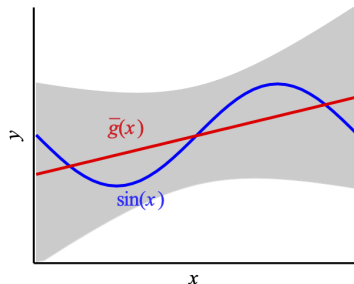
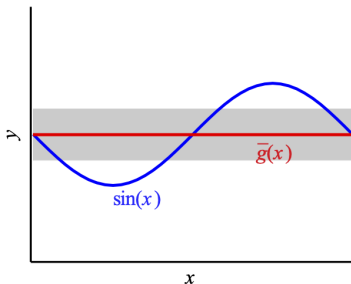
Let's Repeat the Experiment Many Times



- § For each data set $\mathcal{D}$ you get a different $g_n^{\mathcal{D}}$
- § For a fixed x , $g_n^{\mathcal{D}}(x)$ is random value, depending on $\mathcal{D}$

Slide courtesy: Malik Magdon-Ismail

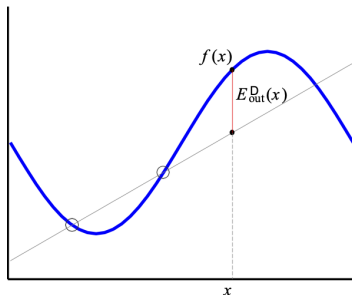
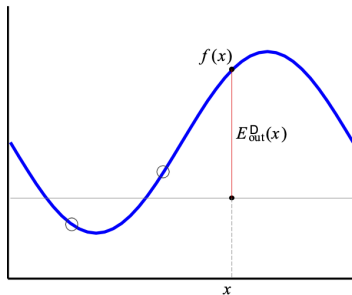
What's Happening on the Average



We can define:

- § $g_n^{\mathcal{D}}(x)$ - Random value, depending on $\mathcal{D}$
- § $\bar{g}(x) = \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)] = \frac{1}{K}(g_n^{\mathcal{D}_1}(x) + g_n^{\mathcal{D}_2}(x) + \dots + g_n^{\mathcal{D}_K}(x))$ - Your average prediction on x
- § $var(x) = \mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - \bar{g}(x))^2] = \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)^2] - \bar{g}(x)^2$ - How variable is your prediction

E_{out} on Test Point x for Data $\mathcal{D}$



§ $E_{out}^{\mathcal{D}}(x) = (g_n^{\mathcal{D}}(x) - f(x))^2$ - Squared error, a random value depending on $\mathcal{D}$

§ $E_{out}(x) = \mathbb{E}_{\mathcal{D}}[E_{out}^{\mathcal{D}}(x)] = \mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - f(x))^2]$ - Expected value of the above random variable

Slide motivation: Malik Magdon-Ismail

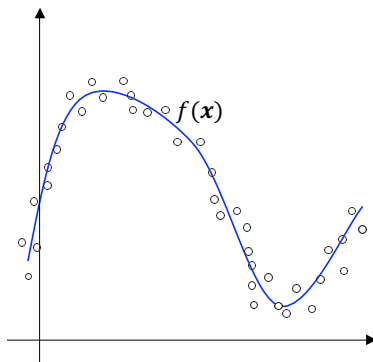
The Bias-Variance Decomposition

$$\begin{aligned}E_{out}(x) &= \mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - f(x))^2] \\&= \mathbb{E}_{\mathcal{D}}[f(x)^2 - 2f(x)g_n^{\mathcal{D}}(x) + g_n^{\mathcal{D}}(x)^2] \\&= f(x)^2 - 2f(x)\mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)] + \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)^2] \\&= f(x)^2 - 2f(x)\bar{g}(x) + \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)^2] \\&= f(x)^2 - \bar{g}(x)^2 + \bar{g}(x)^2 - 2f(x)\bar{g}(x) + \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)^2] \\&= (f(x) - \bar{g}(x))^2 + \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)^2] - \bar{g}(x)^2\end{aligned}$$

$$E_{out}(x) = \underbrace{(f(x) - \bar{g}(x))^2}_{\text{Bias}} + \underbrace{\mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - \bar{g}(x))^2]}_{\text{Variance}}$$

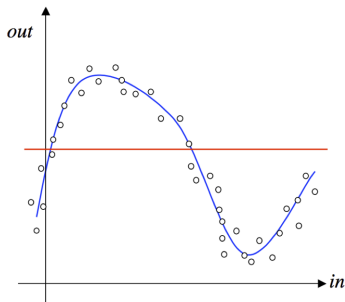
Bias-Variance to Overfitting-Underfitting

- § Suppose the true underlying function is $f(x)$
- § But the observations (data points) are noisy *i.e.*, $f(x) + \epsilon$



The Extreme Cases of Bias and Variance - Underfitting

- § A good way to understand the concepts of bias and variance is by considering the two extreme cases of what a model might learn.
- § Suppose the model is lazy and just produces the same constant output whatever training data we give it, *i.e.*, $g_n^D(x) = c$, then,



Ignore the data $\Rightarrow$

Big approximation error (high bias)

No variation between data sets (no variance)

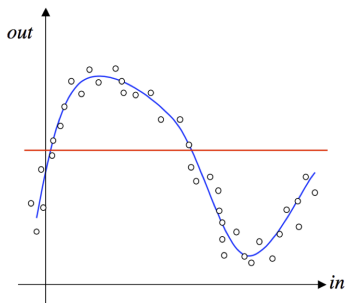
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§ Suppose the model is lazy and just produces the same constant output whatever training data we give it, *i.e.*, $g_n^{\mathcal{D}}(x) = c$, then,

$$\begin{aligned} E_{out}(x) &= (f(x) - \bar{g}(x))^2 + \mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - \bar{g}(x))^2] \\ &= (f(x) - c)^2 + \mathbb{E}_{\mathcal{D}}[(c - c)^2] \\ &= (f(x) - c)^2 + 0 \end{aligned}$$

§ In this case the variance term will be zero, but the bias will be large, because the network has made no attempt to fit the data. We say we have extreme under-fitting.



Ignore the data $\Rightarrow$

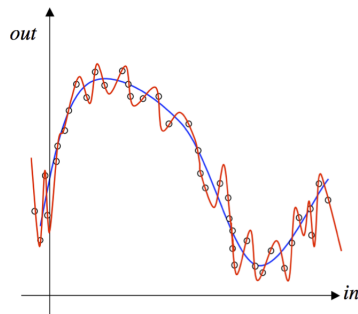
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The Extreme Cases of Bias and Variance - Overfitting

§ On the other hand, suppose the neural network is very hard working and makes sure that it exactly fits every data point, *i.e.*, $g_n^{\mathcal{D}}(x) = f(x) + \epsilon$, then,

$$\bar{g}(x) = \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)] = \mathbb{E}_{\mathcal{D}}[f(x) + \epsilon] = \mathbb{E}_{\mathcal{D}}[f(x)] + 0 = f(x)$$



Fit every data point $\Rightarrow$

No approximation error (zero bias)

Variation between data sets (high variance)

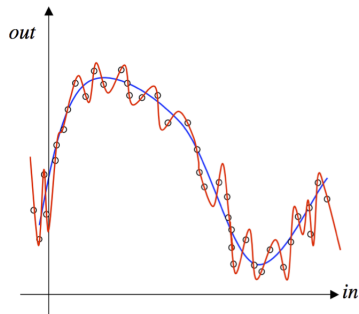
The Extreme Cases of Bias and Variance - Overfitting

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$$\begin{aligned} \S \quad \bar{g}(x) &= \mathbb{E}_{\mathcal{D}}[g_n^{\mathcal{D}}(x)] = \mathbb{E}_{\mathcal{D}}[f(x) + \epsilon] = \\ &\mathbb{E}_{\mathcal{D}}[f(x)] + 0 = f(x) \end{aligned}$$

$$\begin{aligned} E_{out}(x) &= (f(x) - \bar{g}(x))^2 + \mathbb{E}_{\mathcal{D}}[(g_n^{\mathcal{D}}(x) - \bar{g}(x))^2] \\ &= (f(x) - f(x))^2 + \mathbb{E}_{\mathcal{D}}[(f(x) + \epsilon - f(x))^2] \\ &= 0 + \mathbb{E}_{\mathcal{D}}[\epsilon^2] \end{aligned}$$

§ In this case the bias term will be zero, but the variance will a positive value. Because the network has made every attempt to fit the data, we say we have over-fitting.



Fit every data point $\Rightarrow$

No approximation error (zero bias)

Variation between data sets (high variance)

Bias-Variance Trade-off

